

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned end-to-end design, development, and systems integration of cross-platform backend for 100+ users, building REST API-based data collection workflows and enabling reliable downstream analytics and reporting
- Designed and optimized PostgreSQL schema, indexing strategies, and query execution plans for high-volume datasets, improving query performance by 38% to support low-latency API responses and downstream integrations
- Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by **43%**
- Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by **41%**
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection
- Designed production-grade REST APIs using FastAPI, implementing validation, error handling, monitoring, and alerting, and built automated deployment workflows to support reliable systems integration across distributed operational environments
- Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable, microservices-based AI system using FastAPI and PostgreSQL, designing REST API integrations and automated data workflows for production-grade enterprise applications
- Designed fault-tolerant ETL pipelines with fail-safe mechanisms and error handling to process large datasets for low-latency retrieval, ensuring system reliability and data protection across production environments
- Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements in production environments
- Developed backend services enabling real-time, high-performance querying of unstructured data across distributed system architectures

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services and decision-making insights
- Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance issues in distributed systems environments
- Developed backend services using Git-based CI/CD workflows, maintaining technical documentation and architecture diagrams to support reliable deployment across distributed, high-availability environments

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Science Intern

Mumbai, IN

- Developed and maintained 20+ Power BI dashboards across Finance and Operations for enterprise clients, using SQL data extraction and DAX to build reports and dashboards supporting operational KPI tracking and decision-making
- Automated recurring **Excel**-based reporting workflows using **VBA**, implementing data validation, transformation, and scheduling logic, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led a cross-functional team of 6 engineers and designers to deliver 10+ product enhancements, translating user requirements and analytics
- Utilized gameplay telemetry and user feedback data to prioritize engineering efforts, improving delivery efficiency and product quality

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation
- Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency to improve model performance
- Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers